



CLOUD COMPUTING LAB

Lab 04



MARCH 12, 2026

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01-131232-013

Containerization and Deployment using Docker

Lab Objective:

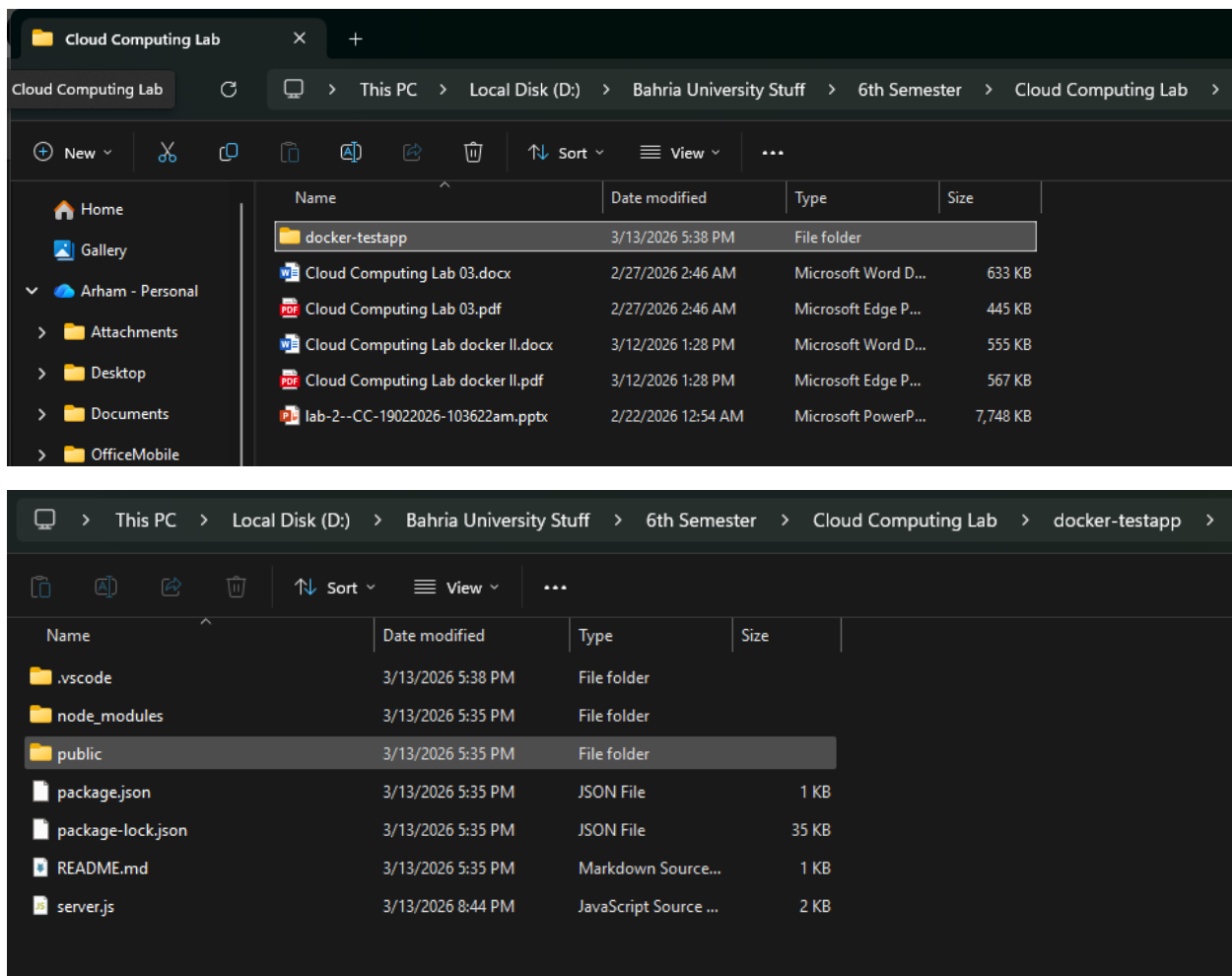
The objective of this lab is to introduce you to containerization technologies using Docker and to understand how applications can be packaged, deployed, and distributed using containers

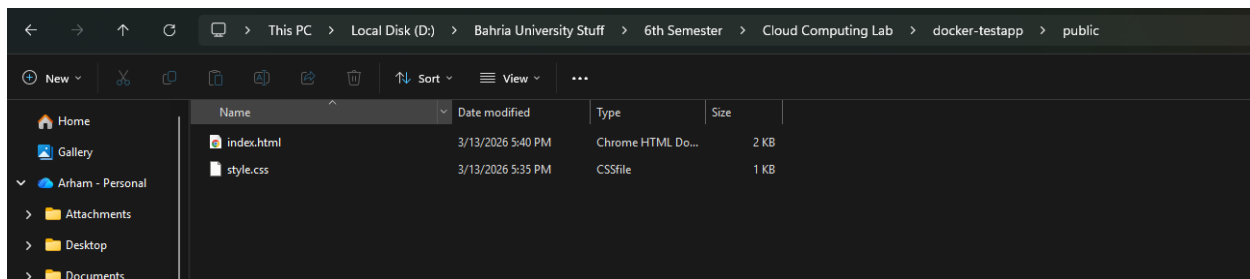
GitHub Repository Link:

<https://github.com/arhamzargham/CloudComputingLab04-Containerization-and-Deployment-using-Docker.git>

Task 1: Developing with Docker

Screenshot of project folder



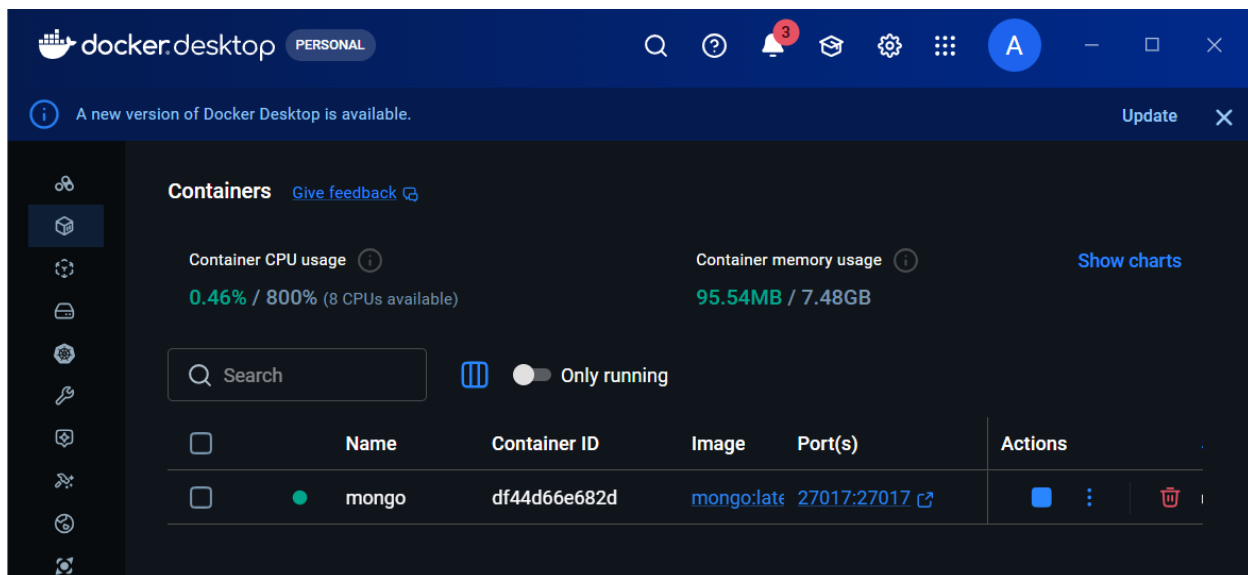


Screenshot of running container

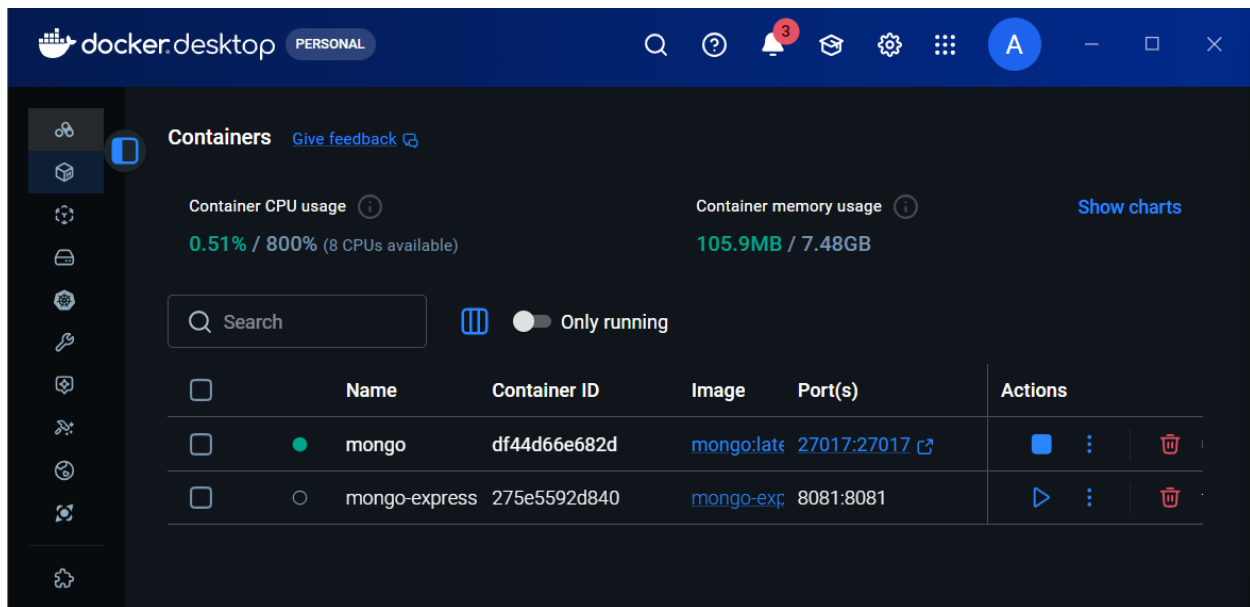
```
C:\Users\Laptop inn>docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
3979eadd1884        bridge             bridge              local
e260cc9408a6        host               host                local
6e0d9b3c1978        none               null                local

C:\Users\Laptop inn>docker network create mongo-network
befd4e4738f26f822261a25d744299aa84818c51a4f6a52d54c38c3a84
6d4993

C:\Users\Laptop inn>docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
3979eadd1884        bridge             bridge              local
e260cc9408a6        host               host                local
befd4e4738f2        mongo-network      bridge              local
6e0d9b3c1978        none               null                local
```



```
C:\Users\Laptop inn>docker run -d^
More? -p 27017:27017 ^
More? --name mongo ^
More? --network mongo-network^
More? -e MONGO_INIT_ROOT_USERNAME=admin^
More? -e MONGO_INIT_ROOT_PASSWORD=qwerty^
More? mongo:latest
Unable to find image 'mongo:latest' locally
latest: Pulling from library/mongo
f97b55066391: Pull complete
8d23bbf2b216: Pull complete
fbfd783f7a59: Pull complete
cea2e6a72e6b: Pull complete
e46eb2190ffc: Pull complete
e79e499203c9: Pull complete
01d7766a2e4a: Pull complete
427c118b2079: Pull complete
4eeb70347fbd: Download complete
bae1425230be: Download complete
Digest: sha256:3a7947f3211e70c045ebff7e94293298bc3fa4e08e5
cfcb8cb32ca0d06e06635
Status: Downloaded newer image for mongo:latest
df44d66e682dd723265659b19fa520761745f68ba683dabe2850c07c75
b9df13
```

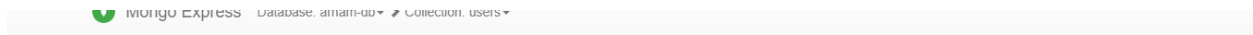


```
C:\Users\Laptop inn>docker run -d ^
More? --name mongo-express ^
More? -p 8081:8081 ^
More? --network mongo-network ^
More? -e ME_CONFIG_MONGODB_ADMINUSERNAME=admin ^
More? -e ME_CONFIG_MONGODB_ADMINPASSWORD=qwerty ^
More? -e ME_CONFIG_MONGODB_URL="mongodb://admin:qwerty@mon
go:27017" ^
More? mongo-express
Unable to find image 'mongo-express:latest' locally
latest: Pulling from library/mongo-express
88f4f8a6bc8d: Pull complete
619be1103602: Pull complete
7e9a007eb24b: Pull complete
0bf3571b6cd7: Pull complete
9f7f59574f7d: Pull complete
45b24ec126f9: Pull complete
d8305ae32c95: Pull complete
5189255e31c8: Pull complete
337574bd0f5e: Download complete
99ef4db52d2b: Download complete
Digest: sha256:1b23d7976f0210dbec74045c209e52fbb26d29b2e87
3d6c6fa3d3f0ae32c2a64
Status: Downloaded newer image for mongo-express:latest
275e5592d84013f6ff33076d97c5a843e8f048be298b2ab3381b0269bf
c01a23
```



Hi student, Welcome to Bahria University!

You can sign up to the platform to access all important notes & lectures.



Viewing Collection: users

Document deleted! _id: 69b452db89effc8ece192b79

[New Document](#) [New Index](#)

Simple

Advanced

Delete all 2 documents retrieved

_id	email	username	password
69b44a11536db01621f8c286	arham@gmail.com	Arham Jan	secret
69b452d789effc8ece192b78	arham@gmail.com	Arham Jan	zargham100504

Rename Collection

```
Pretty-print ☒
[
  {
    "_id": "69b44a11536db01621f8c286",
    "email": "arham@gmail.com",
    "username": "Arham Jan",
    "password": "secret"
  },
  {
    "_id": "69b452d789effc8ece192b78",
    "email": "arham@gmail.com",
    "username": "Arham Jan",
    "password": "zargham100504"
  }
]
```

Short explanation (5–6 lines)

In this task, a Node.js application was developed and executed inside an isolated Docker container. Docker packages the application code, its dependencies, and the required Node.js runtime environment into a single, standardized unit. When executed, the Docker engine runs this image as a container, providing a dedicated filesystem and network space that is completely separate from the host operating system. This isolation ensures that the application code interacts smoothly and predictably with the Docker runtime, eliminating environment-specific inconsistencies. Finally, by mapping the host machine's port to the container's port, we successfully accessed and ran the containerized Node.js service locally.

Task 2: Docker Compose

Screenshot of docker-compose file

```
1 version:"3.8"
2 services:
3   mongo:
4     image:mongo
5     ports:
6       - "27017:27017"
7     environments:
8       MONGO_INITDB_ROOT_USERNAME: admin
9       MONGO_INITDB_ROOT_PASSWORD: qwerty
10
11   mongo-express:
12     image:mongo-express
13     ports:
14       - "8081:8081"
15     environments:
16       ME_CONFIG_MONGODB_ADMINUSERNAME: admin
17       ME_CONFIG_MONGODB_ADMINPASSWORD: qwerty
18       ME_CONFIG_MONGODB_URL: mongodb://admin:qwerty@mongo:27017/
19
20
```

Containers [Give feedback](#)



Your running containers show up here

A container is an isolated environment for your code

 ?

What is a container?
5 mins

```
1 FROM node
2 RUN mkdir -p
3 WORKDIR /app
4 COPY package
```

How do I run a container?
6 mins

[View more in the Learning center](#)

Screenshot of running containers

The screenshot displays the Docker Desktop interface. At the top, the 'Local' tab is active, showing a storage usage bar at 1.18 GB / 3.1 GB in use and 2 images. A search bar and a 'Delete' button are visible. Below this is a table of images:

Name	Tag	Image ID	Created	Size	Actions
mongo-express	latest	1b23d7976f02	2 years ago	286.79 MB	[Play] [More] [Delete]
mongo	latest	3a7947f3211e	24 days ago	1.29 GB	[Play] [More] [Delete]

Below the table, it indicates 'Selected 1 of 2'. The 'Terminal' tab is active, showing the output of the command `docker compose -f mongodb.yaml up -d`. The output shows that the images were pulled and the containers were created successfully.

```
PS D:\Bahria University Stuff\6th Semester\Cloud Computing Lab\docker-testapp> docker compose -f mongodb.yaml up -d
time="2026-03-14T00:11:46+05:00" level=warning msg="D:\Bahria University Stuff\6th Semester\Cloud Computing Lab\docker-testapp\mongodb.yaml
: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 25/25
✓ Image mongo Pulled 141.1s
✓ Image mongo-express Pulled 73.0ss
✓ Network docker-testapp_default Created 0.3s
✓ Container docker-testapp-mongo-1 Created 2.6s
✓ Container docker-testapp-mongo-express-1 Created 2.5s
PS D:\Bahria University Stuff\6th Semester\Cloud Computing Lab\docker-testapp>
```

The bottom status bar shows system resources: RAM 2.46 GB, CPU 22.39%, and Disk 3.94 GB used (limit 1006.85 GB). There is also a 'Terminal' button and an 'Update available' notification.

Explanation of how Docker Compose works

Docker Compose is an orchestration tool designed to simplify the process of defining, managing, and running multi-container applications. Instead of manually executing multiple long `docker run` commands to start individual containers and connect them to shared networks, Docker Compose allows you to declare your entire application stack—including services, ports, environments, and networks—within a single, easy-to-read `mongodb.yaml` file. By executing a single command like `docker compose up`, the tool automatically reads this configuration and spins up all interconnected containers simultaneously. In this task, it allowed the MongoDB database and Mongo Express to launch together and communicate seamlessly.

Task 3: Dockerizing the Application

Screenshot of Dockerfile

```
Dockerfile X
Dockerfile
1 FROM node
2
3 ENV MONGO_DB_USERNAME=admin \
4     MONGO_DB_PWD=qwerty \
5
6
7 RUN mkdir -p DOCKER-TESTAPP
8
9 COPY . /DOCKER-TESTAPP
10
11 CMD ["node", "/DOCKER-TESTAPP/server.js"]
```

Screenshot of Docker image build

The screenshot shows the Docker Desktop interface. On the left is a sidebar with navigation options: Ask Gordon, Containers, Images (selected), Volumes, Kubernetes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main area is titled 'Images' and shows a progress bar for 'Local' images (1.18 GB / 3.1 GB in use) and 'My Hub' images (3 Images). Below the progress bar is a search bar and a table of images.

	Name	Tag	Image ID	Created	Size	Actions
☒	mongo-express	latest	1b23d7976f02	2 years ago	286.79 MB	
☐	mongo	latest	3a7947f3211e	24 days ago	1.29 GB	
☐	testapp	1.0	c1a0d44298c5	2 seconds ago	1.63 GB	

Below the table is a 'Terminal' section showing the output of a 'docker build' command. The output indicates that the build was successful, creating the 'testapp:1.0' image. The terminal text is as follows:

```
PS D:\Bahria University Stuff\6th Semester\Cloud Computing Lab\docker-testapp> docker build -t testapp:1.0 .
[+] Building 36.2s (5/8)
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 214B
=> [internal] load metadata for docker.io/library/node:latest
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
[+] Building 145.9s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 214B
=> [internal] load metadata for docker.io/library/node:latest
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
```

At the bottom of the interface, there is a status bar showing 'Engine running', 'RAM 2.87 GB', 'CPU 0.50%', and 'Disk: 5.49 GB used (limit 1006.85 GB)'. There are also buttons for 'Terminal' and 'Update available'.

Containers

Give feedback

Container CPU usage

0.99% / 800% (8 CPUs available)

Container memory usage

170.53MB / 7.48GB

Show charts

Search

Only show running containers

		Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐		awesome_curran	3de1e76dbb62	testapp:1.0		0%	22 seconds ago	
☐	>	docker-testapp	-	-	-	1.21%	25 minutes ago	

Terminal

+ - x

PS D:\Bahria University Stuff\6th Semester\Cloud Computing Lab\docker-testapp> docker run testapp:1.0

server running on port 5050

Search

Windo...

Windo...

Windo...

Screenshot of running application

Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\Laptop inn> cd "D:\Bahria University Stuff\6th Semester\Cloud Computing Lab\docker-testapp"

PS D:\Bahria University Stuff\6th Semester\Cloud Computing Lab\docker-testapp> docker run -it testapp:1.0 bash

root@3b24ef184563:/#

Search

Windo...

Windo...

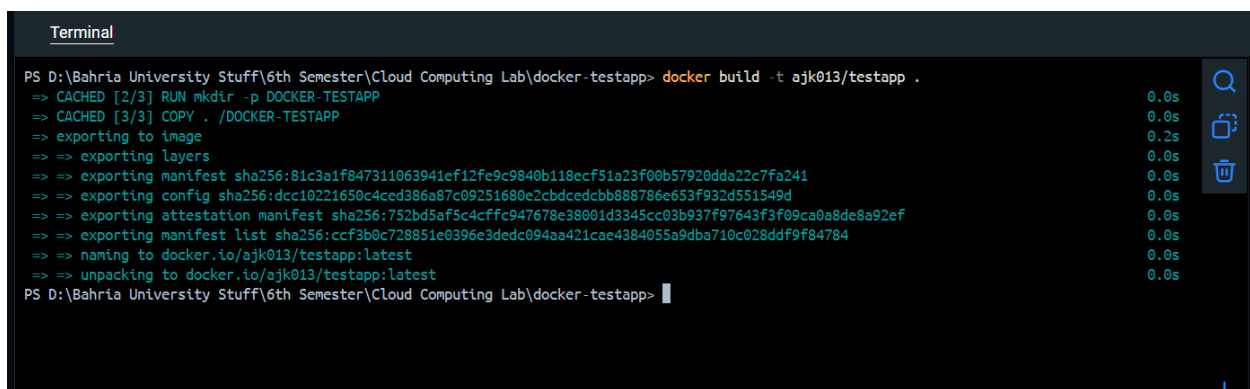
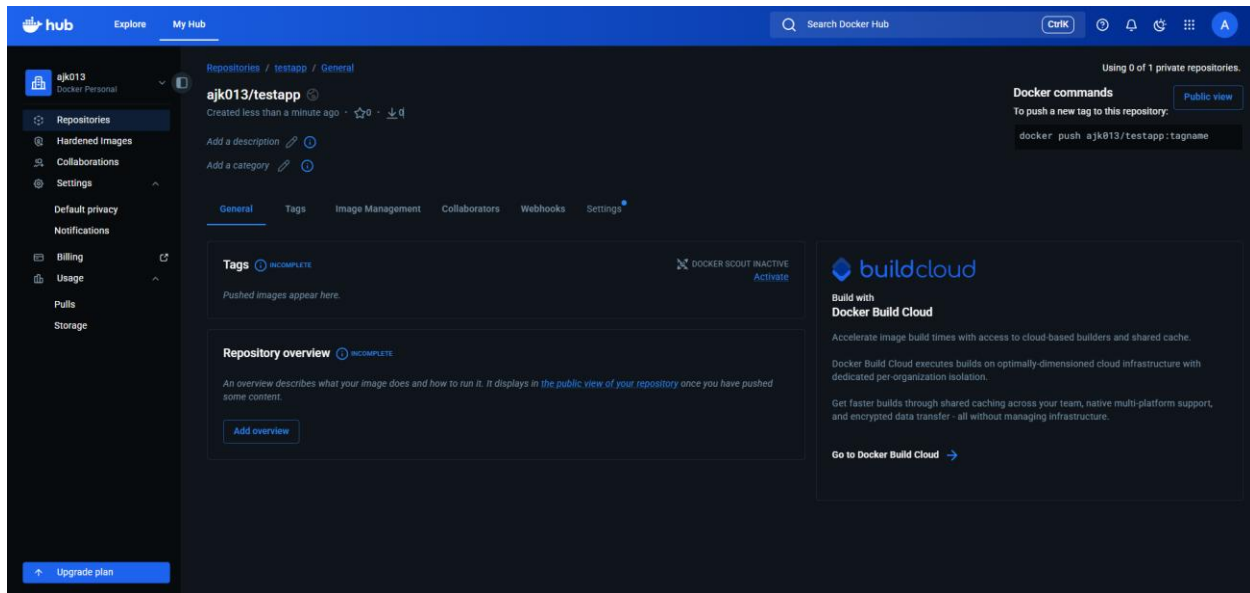
Windo...

RAM 2.88 GB CPU 2.25% Disk: 5.49 GB used (limit 1006.85 GB)

>_ Terminal Update available

Task 4: Publishing Image to Docker Hub

Screenshot of Docker Hub repository



Screenshot of pushed image



The screenshot shows the Docker Hub interface for the repository **ajk013/testapp**. The left sidebar contains navigation links: Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, Pulls, and Storage. The main content area shows the repository details, including the name **ajk013/testapp**, the last push time (less than a minute ago), and the repository size (390.6 MB). Below this, there are tabs for General, Tags, Image Management, Collaborators, Webhooks, and Settings. The **Tags** tab is active, showing a table with one tag: **latest**. The table has columns for Tag, OS, Type, Pulled, and Pushed. The **latest** tag is of type **Image**, pulled less than 1 day ago, and pushed less than a minute ago. Below the table, there is a **Repository overview** section with an **INCOMPLETE** status and a button to **Add overview**.

Repositories / testapp / General

ajk013/testapp ⓘ

Last pushed less than a minute ago · Repository size: 390.6 MB · ☆ 0 · ↓ 0

Add a description ⓘ ⓘ

Add a category ⓘ ⓘ

General Tags Image Management Collaborators Webhooks Settings

Tags ⓘ DOCKER SCOUT INACTIVE [Activate](#)

This repository contains 0 tag(s).

Tag	OS	Type	Pulled	Pushed
latest	linux	Image	less than 1 day	less than a minute

[See all](#)

Repository overview ⓘ INCOMPLETE

An overview describes what your image does and how to run it. It displays in the [public view of your repository](#) once you have pushed some content.

[Add overview](#)

[Upgrade plan](#)

Docker Hub link to the image

<https://hub.docker.com/repository/docker/ajk013/testapp>

Docker Pull Command

```
docker pull ajk013/testapp:latest
```